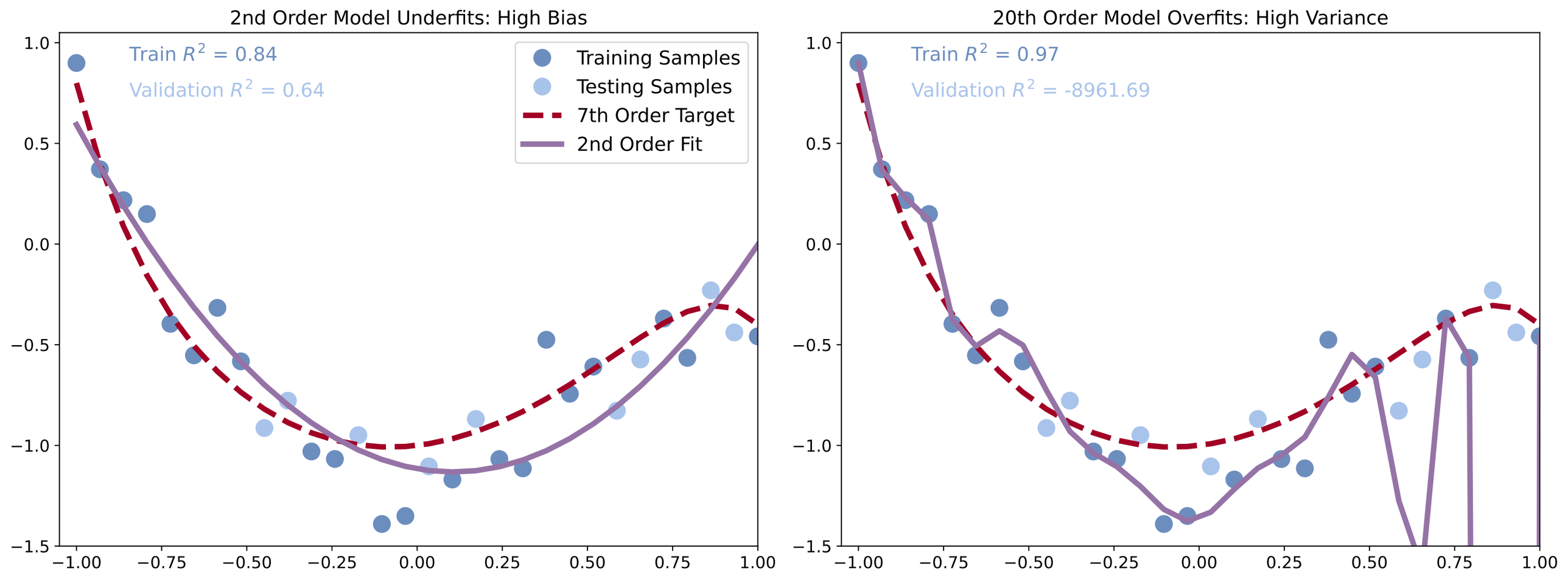
2nd Order Model Underfits: High Bias
Train $R^2$ = 0.84
Validation $R^2$ = 0.64
Training Samples
Testing Samples
7th Order Target
2nd Order Fit
20th Order Model Overfits: High Variance
Train $R^2$ = 0.97
Validation $R^2$ = -8961.69
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00